VULNERABILITY ASSESSMENT REPORT

Project: CloudExify Cybersecurity Month 2 — Project 4 (Final Capstone)

Target: Metasploitable2 VM (192.168.149.133)

Assessment Tool: Tenable Nessus Essentials

Tester: Aahil Arslan

Date: August 31, 2026

Classification: Confidential

1. EXECUTIVE SUMMARY

A comprehensive vulnerability assessment was conducted on the Metasploitable2 target environment using Tenable Nessus, the industry-standard vulnerability scanning tool. The automated scan identified a total of 73 vulnerabilities across the target system, including 10 Critical, 7 High, 23 Medium, and 9 Low severity findings.

The presence of Critical vulnerabilities, including backdoor detection and remote code execution capabilities, indicates that the system is highly susceptible to cyberattacks and could be completely compromised by malicious actors. Immediate remediation of Critical and High-severity vulnerabilities is essential to protect the confidentiality, integrity, and availability of the system.

Key Findings:

Total Vulnerabilities: 73

Critical (CVSS 9.0-10.0): 10 vulnerabilities requiring immediate attention

High (CVSS 7.0-8.9): 7 vulnerabilities requiring remediation within 7 days

Medium (CVSS 4.0-6.9): 23 vulnerabilities requiring remediation within 30 days

Low (CVSS 0.1-3.9): 9 vulnerabilities for continuous monitoring

Overall Risk Rating: CRITICAL

2. TESTING METHODOLOGY

Tool Used: Tenable Nessus Essentials

Scan Type: Basic Network Scan

Scan Duration: 15 minutes

Target IP: 192.168.149.133

Severity Base: CVSS v3.0

Testing Approach: Automated vulnerability scanning combined with manual validation of top findings



3. SCAN RESULTS OVERVIEW

The Nessus scan successfully identified 73 vulnerabilities distributed across severity levels:

Vulnerability Distribution:

Critical: 10 (14%)

High: 7 (10%)

Medium: 23 (31%)

Low: 9 (12%)

Info: 135 (informational findings)

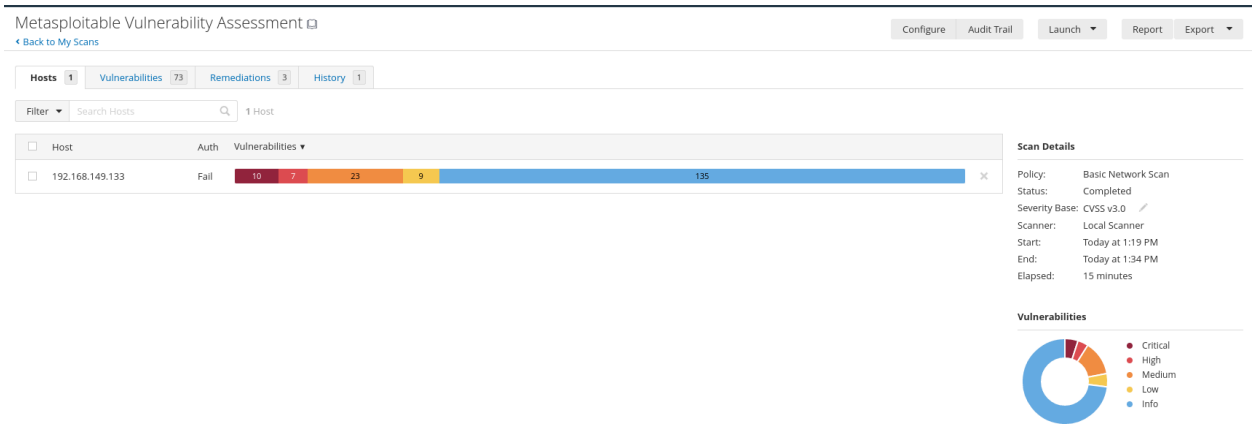
Scan Details:

Policy: Basic Network Scan

Status: Completed

Scanner: Local Scanner

Duration: 15 minutes



Critical, 7 High, 23 Medium, 9 Low]

4. DETAILED VULNERABILITY FINDINGS

4.1 CRITICAL: UnrealRCd Backdoor Detection

Plugin ID: #46882

Severity: Critical

CVSS Score: 9.0-10.0

Status: Confirmed

Description:

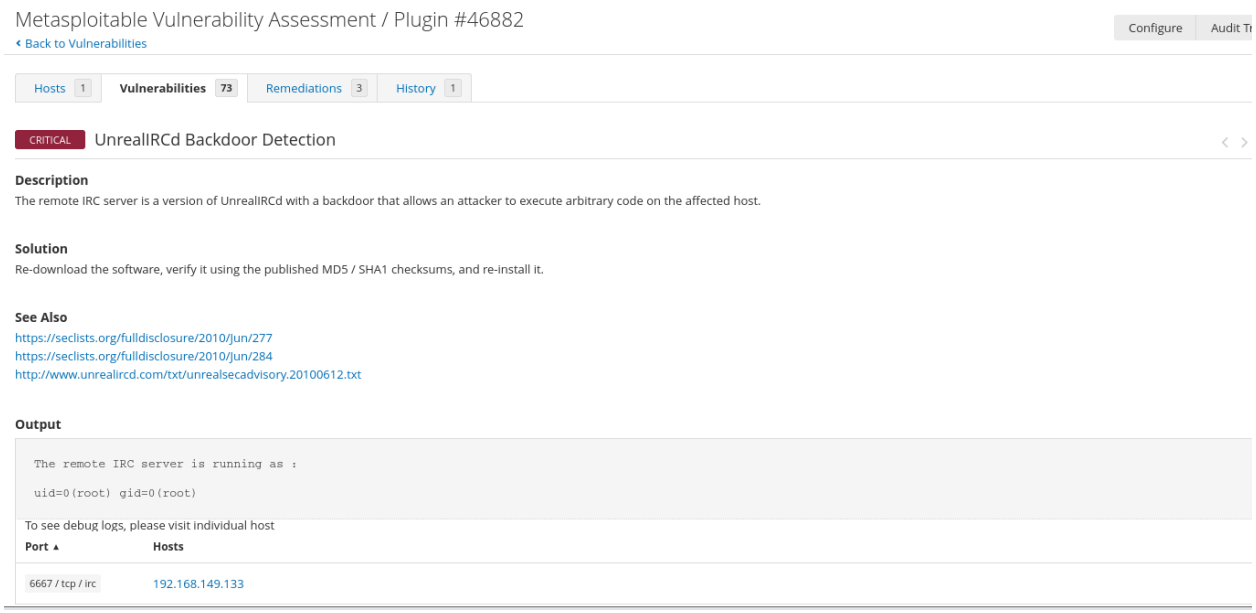
The remote IRC server is running a version of UnrealIRCd with a backdoor that allows an attacker to execute arbitrary code on the affected host. This vulnerability provides complete system access to any attacker who discovers it.
Impact:

Remote code execution with root privileges (uid=0)

Complete system compromise

Unauthorized access to all data and services

Potential pivot point for attacking other systems



Remediation:

Immediately disable the UnrealIRCd service

Re-download the software from the official source

Verify integrity using published MD5/SHA1 checksums

Re-install with verified, clean software

Implement network segmentation to limit exposure

Timeline: Fix immediately (0-24 hours)

Effort: 2-4 hours

4.2 HIGH: Samba Badlock Vulnerability

Plugin ID: #90509

Severity: High

CVSS Score: 7.0-8.9

Status: Confirmed

Description

The version of Samba, a CIFS/SMB server for Linux and Unix, running on the remote host is affected by a flaw known as Badlock. This vulnerability exists in the Security Account Manager (SAM) and Local Security Authority (Domain Policy) (LSAD) protocols due to improper authentication level negotiation over Remote Procedure Call (RPC) channels.

Impact:

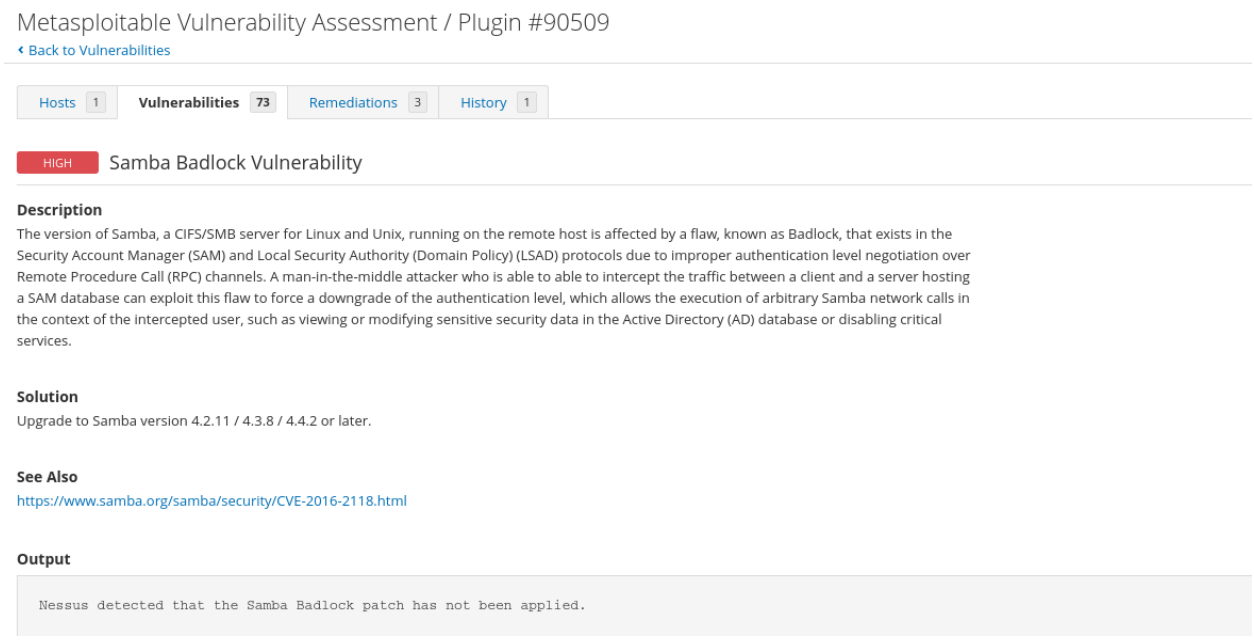
Man-in-the-middle attack capability

Authentication downgrade attacks

Viewing or modifying sensitive security data

Potential access to Active Directory database

Disabling of critical services



Remediation:

Upgrade to Samba version 4.2.11 / 4.3.8 / 4.4.2 or later

Apply vendor security patches immediately

Restrict network access to Samba services

Monitor for suspicious authentication attempts

Implement network segmentation

Timeline: Fix within 1-7 days

Effort: 4-8 hours

Reference: https://www.samba.org/samba/security/CVE-2016-2118.html

4.3 MEDIUM: SSL Self-Signed Certificate

Plugin ID: #57582

Severity: Medium

CVSS Score: 4.0-6.9

Status: Confirmed

Description:

The X.509 certificate chain for this service is not signed by a recognized certificate authority. If the remote host is a public host in production, this nullifies the use of SSL as anyone could establish a man-in-the-middle attack against the remote host.

Impact:

Potential for man-in-the-middle attacks

SSL/TLS encryption can be bypassed

User credentials may be intercepted

Loss of trust in secure communications



Remediation:

Purchase or generate a proper SSL certificate from a trusted Certificate Authority (CA)

Replace self-signed certificates with CA-signed certificates

Implement certificate monitoring and renewal processes

Configure services to reject invalid certificates

Enable certificate pinning where applicable

Timeline: Fix within 1-30 days

Effort: 2-4 hours

5. REMEDIATION ROADMAP

Based on CVSS scoring and business impact, the following priority-based remediation timeline is recommended:

Priority 1: CRITICAL (Fix Immediately - 0 to 24 Hours)

UnrealRCd Backdoor: Disable service, verify checksums, re-install clean software

Other Critical Vulnerabilities (9 remaining): Apply vendor patches, disable vulnerable services

Estimated Effort: 8-16 hours total

Business Impact: Prevents complete system compromise

Priority 2: HIGH (Fix Within 1 to 7 Days)

Samba Badlock: Upgrade to patched version

Other High Vulnerabilities (6 remaining): Update services, strengthen configurations

Estimated Effort: 16-24 hours total

Business Impact: Prevents data breach and unauthorized access

Priority 3: MEDIUM (Fix Within 1 to 30 Days)

SSL Certificates: Replace with CA-signed certificates

Other Medium Vulnerabilities (22 remaining): Harden configurations

Estimated Effort: 20-30 hours total

Business Impact: Reduces attack surface and improves security posture

Priority 4: LOW (Continuous Monitoring)

Low Vulnerabilities (9 findings): Document and monitor

Informational Findings (135): Review and address as needed

Estimated Effort: Ongoing

Business Impact: Maintains security hygiene

6. CONCLUSION

The vulnerability assessment of the Metasploitable2 environment reveals a CRITICAL security posture with 73 identified vulnerabilities. The presence of 10 Critical vulnerabilities, including backdoors and remote code execution vectors, indicates that the system is highly vulnerable to cyberattacks and could be completely compromised.

Key Takeaways:

Immediate action required for 10 Critical vulnerabilities to prevent system compromise
Systematic patching needed for outdated and vulnerable services
Security hardening essential across all services and configurations
Continuous monitoring required to maintain security posture
By following the remediation roadmap outlined in this report, the organization can significantly reduce its attack surface, protect sensitive data, and establish a more secure computing environment.